MILENA COSTA



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LinkedIn · GitHub · Website

No current or future visa sponsorship required

RESEARCH PROFILE

I study how social contexts shape individual well-being, combining psychological theory with statistical methods on population-scale register and survey data. My work connects micro-level processes (e.g., how ties across life domains relate to trust) to population-level patterns, with the aim of informing social policy. I have research experience in psychometrics, multilevel modelling, and network analysis.

EDUCATION

MSc Statistics & Data Science – Leiden University September 2024 – July 2026

Specialisation in Official Statistics (EMOS)

Thesis (8.0): The Contexts that Bind Us: Networks, Multiplexity, and Embeddedness in Institutional Domains

BSc Psychology – Maastricht University August 2021 – July 2024

Thesis (8.5): Expanding the Job Demands-Resources Model: The Work Dynamics Model (work–life balance)

PUBLICATIONS & PRESENTATIONS

- Costa, M., de Jonge, E., & van Klingeren, F. (2026). Cross-domain embeddedness and generalised trust in the Netherlands. CBS Discussion Paper. *Manuscript in preparation.*
- Costa, M. (2026, November). The Contexts That Bind Us: Networks, Multiplexity, and Embeddedness in Institutional Domains [Conference presentation]. ODISSEI Conference, Utrecht, The Netherlands.
- Also presented at CBS Seminar (The Hague, 2026) and EMOS Symposium (The Hague, 2026).

RESEARCH EXPERIENCE

Master's Thesis Intern – Centraal Bureau voor de Statistiek (CBS) January 2026 – June 2026

- Developed an adjusted multiplexity measure to correct for administrative overlap between family and household registers, reducing estimated prevalence from 81.3% to 23.8%.
- Linked a whole-population multilayer network (∼17 million Dutch residents) to survey data (N = 28,483) to examine how the organisation of social ties across institutional domains relates to generalised trust, a key indicator of social cohesion.
- Built a reproducible R pipeline (data.table, DuckDB, Arrow) for large-scale network computation.

Statistical Consulting Project – Leiden University September 2025 – January 2026

- Applied multilevel GLMMs to assess how peer norms shaped children's pro-environmental effort choices; identified ceiling effects that informed task-design recommendations for the client.
- Developed a size-adjusted network density measure of classroom cohesion from peer-nomination data.

Research Project – Vrije Universiteit Amsterdam September 2023 – November 2023

- Evaluated a brief self-report well-being scale for workplace settings, examining its factor structure, reliability, and construct validity.

SKILLS

Statistical methods: Psychometrics (scale validation, factor analysis, IRT, DIF), longitudinal and mixed-effects modelling, GLMs, SEM, survey methodology, statistical learning, Bayesian methods, network analysis

Programming: R (psych, mirt, lavaan, lme4, igraph, data.table, DuckDB), Python, Quarto, Git/GitHub, LaTeX, SPSS

Languages: Portuguese (native), English (fluent), Dutch (B2+, current university course toward C1), Spanish (B1)

OTHER EXPERIENCE

Volunteer Coordinator – Studium Generale, Maastricht University 2022 – 2024

Coordinated 80+ volunteers across three editions of the Pleasure, Art & Science Festival.

Student Assistant – School of Business and Economics, Maastricht University 2022 – 2024

Evaluated admissions applications against programme criteria, screening 1,000+ candidates across two intake cycles.